

tracking represents only minor variations of the dominant pattern. And it is the dominant pattern that is likely to result in the outcomes we noted in chapter 1 and in the differences in classroom experiences we found in our twenty-five schools. As we have said, this pattern, common to all track systems, contains a number of components. First, the presumption is made that students' related past achievements and potential for learning what will be presented in a particular secondary school course or program of courses can be accurately and fairly measured by tests or the judgments of school people. Second, it is assumed that by dividing students according to the results of whatever criteria or measures are used, groups of similar learners will be formed. Third, it is believed that by grouping like students together individual learning needs will be met and that group instruction is probably adequate for the major part of this task. Fourth, it is thought that, by arranging students in groups so that individual needs can be sufficiently met with common learning goals, instructional activities, and materials, the teaching task is made considerably more manageable for teachers. Fifth, regardless of the process used, students are publicly labeled and categorized according to the school's estimation of their potential as learners. Sixth, the classifications that result from this process are neither neutral nor equal. They form, in fact, a hierarchical system of stratification, with students in the top groups—whether they are labeled “Academic,” “College Preparatory I,” “Gifted,” or “High”—accorded the most value and those in the bottom tracks—whether they are called “Vocational,” “Low,” or “Basic”—accorded the least. And seventh, as will soon become clear as we consider the everyday experiences of students in the twenty-five schools, tracking results in different and *not always equal* educational treatments being given to various groups of students.

These beliefs and features are basic to all tracking systems. They, not the particulars of how specific practices are carried out, are what make the difference in the lives of students. While the differences in tracking systems are fascinating, their chief interest is that they are not likely to make a real difference in tracking effects. There seem to be no *essential* distinctions between various types of tracking. Either schools do or do not track. As we have seen in our twenty-five secondary schools, they almost always do. The results are predictable and remarkably similar.

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The Distribution of Knowledge

As we have seen, there has been a considerable amount of interest in tracking and some scholarly effort spent analyzing it. As a result, we know quite a bit about the outcomes of tracking—what happens to students as a result of being in one or another track, how their academic learning is affected, and what behaviors and attitudes they are likely to exhibit. Other studies have considered the factors that are important in determining who gets placed in which track level. Much of this inquiry has revolved around the question of fairness and has tried to assess the extent to which student placements are based on social class or on “merit.” This question has not yet been resolved to everyone’s satisfaction, because underlying the issue is a whole hotbed of other concerns: the definition of “merit,” the objectivity of standardized tests, and probably the most volatile of all, the relationship between race and scholastic aptitude.

We looked at tracking from a slightly different, but not unrelated, perspective. Consistent with the focus of *A Study of Schooling*, we were interested in learning the content and process of classrooms under tracking systems. We wanted to know what actually goes on in classes at different track levels and how they are similar or different from one another. We wanted to know specific information about what students were being taught, how teachers carried out their instruction, what classroom relationships were like, and how involved students seemed to be in classroom learning. We also wanted to know about what kinds of

student attitudes were characteristic of classrooms—attitudes students had toward themselves, their classrooms, and their schools. Essentially, we wanted to know details about what different kinds of classes were like for students and how students felt about being in them.

These aspects of tracking are not altogether removed from what most other research has considered. In fact, we hoped that by studying tracked classes themselves we could begin to suggest some explanations for why things turn out the way they do for teenagers in different tracks. And because our explanations would revolve around things that go on in schools rather than around characteristics of the students themselves, we could, in turn, suggest things schools might do differently in order to avoid some of the negative results that are related to tracking. But since our interests were somewhat different from most of those who have studied tracking, we needed to approach our study differently as well.

We bypassed, for example, the whole issue of what characteristics of students seem to most influence the decisions about which track they belong in. We did not consider whether the students in our twenty-five schools were placed “fairly.” However, we were very much interested in knowing which kinds of students ended up in each track: rich or poor, white or nonwhite. We were concerned with this because we would be making judgments about the *quality* of the educational experiences in different track levels. And if it turned out, as it did, that there were differences in the quality of educational experiences, we wanted to be able to discuss these differences in terms of educational *equality* for those students who seem to benefit least from schooling—the poor and minorities. Thus we decided to leave the class/merit debate to others and make the bold assertion at the beginning of our work that regardless of how “fair” their placements might appear to be by traditional educational standards, it is never equitable to have any group of students be systematically offered less when it comes to educational quality. By looking at the distribution of various groups of students into track levels next to an assessment of the quality of education offered in those tracks, we could address the issue of equity in a new way. Does the disproportionate placement of poor and minority students in low-track classes mean that they are getting a poorer quality of education in their day-to-day experience in classrooms?

Not surprisingly, then, we were more interested in classrooms as

the focus for our study than we were in individual students. Because we were less concerned with outcomes for individuals than with what students experience in groups, we decided to describe what whole classrooms were like, rather than what the individuals in them were like. Of course, we knew that the characteristics of individuals in classes contribute considerably to the classroom environment. We were most interested, however, in the result of this interaction between individuals and schools—what actually goes on in class. As a result, we used the views of individuals in classrooms collectively as a means of describing the group experience rather than individually as a way of describing students.

Because of this classroom focus, we did not study individual students classified as “academic,” “general,” or “vocational,” as most tracking studies have done. We studied classes that were considered to be at various track levels. We could not use the traditional track labels assigned to students since they did not apply to the academic classes that were a part of the tracking systems at our twenty-five schools. For example, classes for the most highly academic tracks were variously called “advanced placement,” “academically enriched,” “college preparatory level I,” “honors,” or “gifted.” What they all had in common was that they were intended for the highest achievement- or ability-level, college-bound students in the school. We simply grouped together all the tracked classes at similar levels and categorized them as “high,” “average,” or “low.” The administrators, counselors, and teachers at the schools helped us determine which classes belonged in each category.

To discover how the track levels themselves were alike and different, we wanted to study a *representative* group of classes at each level. We wanted to be able to compare tracks fairly without being overinfluenced by class differences that were more related to differences among subjects or to differences among the schools in the number of subjects they tracked than to actual track differences. We settled on the analysis of math and English classes only. We had studied a large number of these classes—297 to be exact. These classes were fairly evenly distributed among the secondary grades. We knew that nearly all of the twenty-five schools tracked in these subjects (twenty-three in English, twenty-four in mathematics). We also knew that nearly every student at our schools took classes in these subjects; most took five or six years of English and at least four years of math. And finally, we recognized that

these two subjects, although both highly academic, differed considerably from each other. We felt that by considering these subject-area differences, we would be able to identify characteristics of classroom processes that were likely to be common to certain track levels in all subjects and those that seemed to be unique to specific subjects. With our decision to study English and math classes, then, we believed that what we found out about classroom processes at various track levels would not be distorted by any particular group of schools, students, or subjects.

We ended up studying 75 high-track classes, 85 average-track classes, and 64 low-track classes that were about evenly divided between math and English. Seventy-five other classes in these two subjects were identified as heterogeneous (or not tracked). These classes proved to be useful, as we will see later, for comparing what went on in tracked and untracked classes.

Throughout this chapter, examples of student and teacher responses will be drawn from high- and low-track classes in many subject areas. These responses are illustrative of the track-level differences we found. However, when conclusions are presented, they are based on the systematic analysis of data about the 299 English and math classes. While it is likely that these conclusions apply to tracking in a general way, it is important to be clear about how they were derived.

Who Goes Where

In chapter 1 we saw clearly that there is a pattern of relationships between students' socioeconomic positions—and important in this is their ethnicity—and their chances of being placed in a particular track level. While there is certainly no automatic placement of poor and minority students in low tracks or of affluent white students in upper tracks, the odds of being assigned into particular tracks are not equal. In virtually every study that has considered this question, poor and minority students have been found in disproportionately large percentages in the bottom groups.

In our study of twenty-five schools, we found this same pattern operating. We were able to examine it directly in two ways—related to student race and ethnicity—and indirectly—related to other socioeconomic characteristics. For one thing, we were able to look closely at the schools with racially mixed populations to determine who got placed in

which track levels at those schools. And second, we were able to look at vocational education programs at all the schools and assess the differences in programs taught to white and nonwhite students. While these two considerations are related directly to race and ethnicity and tracking, they relate indirectly to other socioeconomic status factors as well, for not surprisingly, the minority students in our schools tended to be poorer than the whites. Moreover, the relationships we uncovered were the strongest at schools where the minority students were at the lowest income levels.

Academic Tracking and Race

Our twenty-five schools were very diverse in a number of ways, as we have seen. But one of the most noticeable ways in which they differed was in the racial and ethnic characteristics of the students who attended them. Seven senior highs and six junior highs were attended almost exclusively by white students. These were the schools in the Vista, Crestview, Woodlake, Atwater, Bradford, Euclid, and Dennison communities. The Rosemont schools were Mexican-American, and the Manchester schools black. The other eight schools were racially or ethnically mixed: Fairfield Junior and Senior highs were about half Mexican-American and half white, the Laurel and Palisades schools about half black and half white. Laurel's schools were part of a mixed, although hardly integrated rural community. The blacks at the two Palisades schools were bused in to this affluent white community. The Newport schools, located in a highly diverse metropolitan suburb, were unique among our group. The student population at these schools represented a rich variety of ethnic and racial groups. Slightly less than half of the students at each of the schools were white; the others were Mexican-American, black, or Asian, and a scattering of students were from a number of other distinct ethnic. Together, the thirteen white schools enrolled 10,783 students; the four nonwhite schools, 8,248 students; and the eight mixed schools, 4,287 white and 4,546 nonwhite students.

The relationship between student ethnicity and tracking could be seen at six of our mixed schools: Fairfield, Laurel, and Palisades. At these schools we recorded the race or ethnic background of every student in the classes we studied.¹ By looking at how students from various groups were tracked into the English and math classes at these six schools,

we could check to see if the schools followed the pattern that has been found so consistently in other research.

The white student populations at these six mixed schools ranged from a low of 46 percent to a high of 53 percent with an average for the six of 50 percent. Within these schools, an average of 62 percent of the students in high-track English classes were white, a considerably larger proportion than in the student population as a whole. Only 29 percent of the students in low-track English classes at these six schools were white, a substantially smaller percentage than in the total student population.²

Eight high-track and ten low-track English classes were included in the sample at these six multiracial schools. Of these eighteen classes, fourteen followed the predominant pattern in racial composition, with disproportionately large percentages of white students in high-track classes and of nonwhite students in low-track classes. Of the four classes that did not conform to this racial pattern, three were high-track classes with between 32 and 46 percent white students; the other, a low-track class, had 67 percent white students.

These four classes, however, shared some common characteristics. All four were located in the Palisades community, which, as we have seen, was a middle- to upper-middle-class suburb of a large city. The minority students were middle- and upper-middle-class blacks voluntarily bused to the school. At the other four multiracial schools, the minority populations were considerably less affluent. Additionally, three of these four nonconforming classes were elective subjects—speech, journalism, and creative writing. Only one was a standard language arts class, and that class had the largest white population of any of the three high-track classes (46 percent).

Math classes, too, evidenced this disproportionate allocation of racial groups in track levels. An average of 60 percent of the students in high-track math classes at the six schools were white, compared to only 37 percent of the students in the low-track math classes. As with the English classes, these percentages differed markedly from the percentage of white students in the total population at these multiracial schools.

Six high-track and twelve low-track math classes were studied at these schools. Of these eighteen math classes, only five did not follow the predominant pattern in racial composition—larger percentages of white students in high-track classes and smaller percentages of whites

in low-track classes than in the schools as a whole. Of these five nonconforming classes, two were high-track classes—one with 44 percent white students and one with 29 percent—and three were low-track classes with a percentage of whites ranging from 55 to 65 percent. Like the exceptional English classes, three of these five math classes were located in the Palisades community, which had the more affluent black students.

From the information about these six schools, then, it is clear that in our multiracial schools minority students were found in disproportionately small percentages in high-track classes and in disproportionately large percentages in low-track classes. And, as we have seen, this pattern was most consistently found in schools where minority students were also poor. These findings are consistent with virtually every study that has considered the distribution of poor and minority students among track levels in schools. In academic tracking, then, poor and minority students are most likely to be placed at the lowest levels of the schools' sorting system.

Among vocational classes, the issue of who goes where is more complex. The relationship between students' ethnicity and their placement in vocational programs at our schools will be considered in detail in chapter 8.

Who Learns What

QUESTION: What is the most important thing you have learned or done so far in this class?

We were interested in finding out what students regarded as the most important learnings in their classroom experience. We gave them a considerable amount of empty space on their questionnaires to tell us what they thought. Students in high-track classes tended to write answers like these:

RESPONSES: I've learned to analyze stories that I have read. I can come with an open mind and see each character's point of view. Why she or he responded the way they did, if their response was stupidity or an heroic movement. I like this class because he [the teacher] doesn't put thoughts into your head; he lets you each have a say about the way it happened.

High-track English—senior high

Basic concepts and theories have been most prevalent. We have learned things that are practical without taking away some in-depth studies of the subject.

High-track Science—senior high

Learning political and cultural trends in relation to international and domestic events.

High-track Social Studies—senior high

I have learned a lot about molecules and now am able to reason and figure out more things.

High-track Science—senior high

It teaches you how to do research in a college library.

High-track English—senior high

Learned to analyze famous writings by famous people, and we have learned to understand people's different viewpoints on general ideas.

High-track English—junior high

Things in nature are not always what they appear to be or what seems to be happening is not what really is happening.

High-track Science—senior high

Greek philosophy, Renaissance philosophy, humanities. How to write essays and do term papers. The French Revolution. *HISTORY!*

High-track social studies—junior high

We learned how to do experiments.

High-track science—junior high

I've really learned the whole idea and meaning behind economics and how to apply economics to my life.

The bases of our economic system and the way the business world is.

High-track Vocational Education—senior high

About businesses—corporations, monopolies, oligopolies, etc., and how to start, how they work, how much control they have on the economy—prices, demand, supply, advertising.

We've talked about stocks—bonds and the stock market and about the business in the U.S.A.

High-track Vocational Education—junior high

To infer or apply the *past* ideas to my ideas and finally to the future's ideas.

High-track social studies—junior high

Inductive reasoning.

High-track Math—senior high

Learned about different theories of psychology and about Freud, Fromm, Sullivan and other psychologists.

High-track Social Science—senior high

The most important thing is the way other countries and places govern themselves economically, socially, and politically. Also different philosophers and their theories on government and man and how their theories relate to us and now.

High-track Social Studies—junior high

I have learned things that will get me ready for college entrance examinations. Also, many things on how to write compositions that will help me in college.

High-track English—junior high

The thing we did in class that I enjoyed the most was writing poetry, expressing my ideas. We also had a poet come and read to us.

High-track English—senior high

I have learned quite a deal about peoples of other nations, plus the ideas of creation and evolution, ideas that philosophers have puzzled over for years."

High-track Social Studies—senior high

Many things about chemistry and experiments especially how to prove things that we learn by experiments.

High-track Science—senior high

How to read a classic novel and be able to pull out details, and write a complete and accurate report.

High-track English—senior high

We have learned the basics of the law of relativity, and basics in electronics. The teacher applies these lessons to practical situations.

High-track Science—senior high

Probably the most important thing I've learned is the understanding of the balance between man and his environment.

High-track Science—senior high

We have learned about business deals. We have also learned about contracts.

High-track Vocational Education—senior high

Learned many new mathematical principles and concepts that can be used in a future job.

High-track Math—senior high

Learning to change my thought processes in dealing with higher mathematics and computers.

High-track Math—senior high

How to write successful compositions, how to use certain words and their classifications. What to expect in my later years of schooling.

High-track English—junior high

The most important thing that we have done is to write a formal research paper.

High-track English—senior high

There is no one important thing I have learned. Since each new concept is built on the old ones, everything I learn is important.

High-track Math—senior high

To me, there is not a most important thing I learned in this class. Everything or mostly everything I learn in here is *IMPORTANT*.

High-track English—junior high

I have learned to do what scientists do.

High-track Science—junior high

Students in low-track classes told us the following kinds of things:

How to blow up light bulbs.

Low-track Vocational Education—junior high

Really I have learned nothing. Only my roman numerals. I knew them, but not very good. I could do better in another class.

Low-track Math—junior high

I've learned how to get a better job and how to act when at an interview filling out forms.

Low-track English—junior high

How to ride motorcycles and shoot trap

Low-track Science—senior high

How to cook and keep a clean house. How to sew.

Low-track Vocational Education—junior high

The most important thing I have learned in this class I think is how to write checks and to figure the salary of a worker. Another thing is the tax rate.

Low-track Math—senior high

To be honest, nothing.

Low-track Science—senior high

Nothing outstanding.

Low-track Science—senior high

Nothing I'd use in my later life; it will take a better man than I to comprehend our world.

Low-track Science—senior high

I don't remember.

Low-track Social Studies—junior high

The only thing I've learned is how to flirt with the chicks in class. This class is a big waste of time and effort.

Low-track Science—senior high

I learned that English is boring.

Low-track English—senior high

I have learned just a small amount in this class. I feel that if I was in another class, that I would have a challenge to look forward to each and every time I entered the class. I feel that if I had another teacher I would work better.

Low-track Math—junior high

I can distinguish one type rock from another.

Low-track Science—senior high

To spell words you don't know, to fill out things where you get a job.

Low-track English—junior high

Learned about how to get a job.

Low-track English—junior high

Job applications. Job interviews. Preparation for the above.

Low-track English—junior high

Learned to fill out checks and other banking business.

Low-track English—junior high

Spelling, worksheets.

Low-track Science—junior high

How to sew with a machine and how to fix a machine.

Low-track Vocational Education—junior high

Job training.

Low-track English—junior high

How to do income tax.

Low-track Math—senior high

A few lessons which have not very much to do with history.
(I enjoyed it).

Low-track Social Studies—junior high

Most Americans believe that the school curriculum is fairly standard. From what we remember of our own experiences and what we see represented in the media, we have an impression of sameness. Tenth-grade English at one school seems, with only slight variations here and there, to be tenth-grade English everywhere. This seems to be so much so that we would expect a tenth-grader who moves in the middle of the year from Pittsburgh or Pensacola to Petaluma to slip quite easily into a familiar course of study—a little Shakespeare, some famous short stories, a few Greek myths, lists of vocabulary words from the College Entrance Exams, and guidelines for well-developed paragraphs and short expository themes. The same beliefs hold for most academic subjects. For example, isn't eighth-grade math everywhere a review of basic operations, an introduction to algebraic and geometric concepts, with some practice in graphing and scientific notation and a brief glimpse at function and inequalities? How much could eleventh-grade American history differ from class to class or from place to place? Or ninth-grade introductory biology?

Don't misunderstand, however. We, as a society, have no expectation that all tenth-, or eighth-, or eleventh-graders will finish these classes having *learned* all the same things. We know well that some students are more or less interested than others and that some find it more or less difficult than others do. But most of us do assume that the material itself—facts and concepts to be learned, pieces of knowledge and works of scientific literacy or cultural merit to be appreciated—is at least paraded by everyone as they proceed through school. We assume that *everyone* is at least *exposed*. In our study of twenty-five schools we found these assumptions and beliefs to be unsubstantiated by our observations of what actually went on in classrooms.

One of the particulars we were most interested in finding out about was whether students who were placed in different track levels in sub-

jects had the same opportunities to learn the *content* of those subjects. Were students in different track levels being exposed to the same or similar material? If so, were the differences among tracks merely ones of mode of presentation or pace of instruction? If actual content differences did exist, were they socially or educationally important ones—that is, was what some students were exposed to more highly valued by society than what other students were presented? We also wanted to know whether students in different track levels were experiencing about the same amount of learning time. Were some groups of students getting more instruction than others? Were effective instructional techniques being used more in one track than in another? Did teachers seem to perform better with some groups of students than with others?

We studied each of these questions carefully because we knew that the implications of what we found could be far-reaching. We believe that these issues go to the very heart of the matter of educational equity. For beyond the issue of what schools students have access to is the issue of what knowledge and learning experiences students have access to within those schools. If there are school-based or system-related differences in what students are exposed to, are these differences fair? Do they interfere with our commitment to educational equality?

We have long acknowledged and perhaps even overemphasized the ways in which differences among students influence their learning in school. Cultural and socioeconomic patterns have been carefully studied with an eye toward how those patterns characteristic of poor children, and especially poor and minority children, interfere with their opportunities to achieve in school. We have also given attention to the influence of family characteristics, such as support and encouragement, on school success. Measured aptitude for learning or intelligence has received a huge share of research time and money in the search for explanations of differences in student learning outcomes. All these attributes are alike in that they are seen to reside in the student. They are clearly important in the school-learning process, but they are not factors over which schools have much control. As conceived, there is little school people can do to alter them.

We have not, however, paid so much attention to the role of school opportunities in determining what and how much students learn. For, ultimately, students can learn *in school* only those things that the school exposes them to. And this learning is restricted by the time allotted for

it and the mode of instruction employed. Perhaps this is so obvious that it is clearly understood. I suspect, rather, that it is so obvious that it is usually overlooked as important. But the implications of these simple facts of schooling are tremendous. If schools, perhaps in response to differences students bring with them from home, provide them with different kinds of opportunities to learn, then the schools play an active role in producing differences in what and how much students actually learn. The different educational opportunities schools provide to students become the boundaries within which what different students learn *must* be confined. Further, if these opportunities differ in ways that may be important in influencing children's future opportunities both in and out of school, then the differences in learning that schools help produce have profound social and economic as well as educational consequences for students.

We found in our twenty-five schools that students in some classes had markedly different access to knowledge and learning experiences from students in other classes. In nearly every school, some groups of students experienced what we typically think of as tenth-grade English, eighth-grade math, eleventh-grade history, and ninth-grade science. We found also, again in nearly every school, that other groups of students encountered something quite different. And we found that these differences were directly related to the track level of the classes students were in.

In our study we used several sources of information about the 299 English and math classes to shed light on this question of differences in what was likely to be taught and learned in classes in different track levels. Teachers had compiled packages of materials for us about their classes, including lists of the instructional topics they cover during the year, the skills they teach their students, the textbooks they use, and the ways they evaluate their students' learning. Many teachers also gave us copies of sample lesson plans, worksheets, and tests. The teachers were interviewed, and as part of the interview they were asked to indicate the five most important things they wanted their students to learn during the year with them.

In analyzing all these data we looked for similarities and differences in the content of what students were expected to learn in classes at various track levels. We looked both at the substance of what they were exposed to and at the intellectual processes they were expected to use.

We analyzed these similarities and differences systematically³ and from a particular point of view. We did not assume that all knowledge presented in schools is equally valuable in terms of societal worth, as exchange for future educational, social, and economic opportunities. On the contrary, we began with the recognition that some kinds of knowledge are far more valuable in this way than are others.

We were not thinking, of course, about the value of knowledge in a pure—that is, culture-free—sense. The issue of what is worth knowing in this abstract sense is a question philosophers will continue to grapple with. Nor were we thinking about the value of knowledge in a purely educational sense. Again, it is not clear what kinds of learnings may be better than others in the development of a person who is a learner. This is likely to vary dramatically in groups, even those composed of very similar individuals.

These two issues ignore the social and economic ties attached to learning when it becomes housed in schools. Schools, as social institutions, do far more than impart knowledge and skills to students. They do more than pass on the traditions and values, the folkways and mores of the culture, to the young. Schooling is both more and less than education in the purest sense. It includes as an important function the preparation of youth for future adult roles and for their maintenance of the social structure and organizational patterns of society. And because our social structure is a hierarchical one, with different and fairly specific criteria for entry at various levels, schooling becomes what Joel Spring has called a "sorting machine."⁴ By this he meant that the form and substance of the educative process that occurs in schools also select and certify individuals for adult roles at *particular* levels of the social hierarchy. This sorting process results in part from students' access to socially meaningful knowledge and educational experiences.

We analyzed the differences in the content of classes from this perspective. We wanted to explore whether students in different track levels were systematically given access to knowledge that would point them toward different levels in the social and economic hierarchy.

We found considerable differences in the kinds of knowledge students in various tracks had access to. We found also that these differences were not merely equally valued alternative curricula. Rather than being neutral in this sense, they were differences that could have important implications for the futures of the students involved.

For example, students in high-track English classes were exposed to content that we might call "high-status" knowledge in that it would eventually be required knowledge for those going on to colleges and universities. These students studied standard works of literature, both classic and modern. Some classes traced the historical development of literature, some studied the characteristics of literary genres (the novel, the short story, poetry, the essay), and others analyzed literary elements in these works (symbolism, irony, metaphoric language). Students in these classes were expected to do a great deal of expository writing, both thematic essays and reports of library research. In some classes, too, students were taught to write in particular styles or to learn the conventions of writing in the various literary forms. These students were expected to learn the vocabulary they would encounter on the College Board Entrance (SAT) exams and practice the type of reading comprehension exercises they would find there as well. Some, although not many, of these classes studied language itself, including historical analyses and semantics.

Low-track English classes rarely, if ever, encountered these kinds of knowledge or were expected to learn these kinds of skills. Not only did they not read works of great literature, but we found no evidence of good literature being read to them or even shown to them in the form of films. What literature they did encounter was so-called young-adult fiction—short novels with themes designed to appeal to teenagers (love, growing pains, gang activity) and written at a low level of difficulty. These novels constituted part of the focus of low-track classes on basic literacy skills. Prominent in these classes was the teaching of reading skills, generally by means of workbooks, kits, and reading texts in addition to young-adult fiction. The writing of simple, short narrative paragraphs and the acquisition of standard English usage and functional literacy skills (filling out forms, applying for jobs) were also frequently mentioned as course content in low-track classes.

It is probably not surprising, given the differences in *what* they were learning, that the differences in the intellectual processes expected of students in classes at different levels were substantial. Teachers of the high-track classes reported far more often than others that they had students do activities that demanded critical thinking, problem solving, drawing conclusions, making generalizations, or evaluating or synthesizing knowledge. The learnings in low-track classes, in nearly all cases,

required only simple memory tasks or comprehension. Sometimes low-track students were expected to apply their learnings to new situations, but this kind of thinking was required far less frequently than were memorization and simple understanding.

The teachers of classes intended for "average" students gave us information indicating that the learnings encountered in their classes were somewhere in between the high- and low-track extremes. But it is worth noting that the kinds of knowledge and intellectual skills emphasized in these average English classes were far more like those in the high track than in the low. It is more appropriate to consider these classes as watered-down versions of high-track classes than as a mixture of the other two levels. Low-track classes seemed to be distinctly different.

Math classes followed a similar pattern of differences with one major exception. The knowledge presented in high-track classes in math, as in English, was what we could call "high status"; it was highly valued in the culture and necessary for access to higher education. Topics frequently listed included mathematical *ideas*—concepts about numeration systems, mathematical models, probability, and statistics—as well as computational procedures which became increasingly sophisticated at the higher grades.

In contrast, low-track classes focused grade after grade on basic computational skills and arithmetic facts—multiplication tables and the like. Sometimes included in these classes were simple measurement skills and the conversion of the English system into the metric. Many low-track classes learned practical or consumer math skills as well, especially at the high school level—the calculation of simple and compound interest, depreciation, wages, and so on. Few mathematical ideas as such seemed to be topics of instruction in these classes. In essence, while the content was certainly useful, almost none of it was of the high-status type.

As in the English classes, the average math classes were considerably more like the high-track classes in their content than like the low. And, too, the content of average math classes can be considered a diluted version of that of the high classes. This was especially true at the junior highs and through about grade ten at the senior highs. From that point on in our schools, math was usually no longer a required subject, and only what would be considered high-track classes were offered to those students wishing to go on in math.

Math classes did differ from English classes in the intellectual processes demanded of students in classes at the various track levels. While the topics of math classes differed considerably—and the differences in the conceptual difficulty of these topics is dramatic—students at all levels of math classes were expected to perform about the same kinds of intellectual processes. That is, at all levels, a great deal of memorizing was expected, as was a basic comprehension of facts, concepts, and procedures. Students at all levels were also expected to apply their learnings to new situations—whether it was the application of division facts to the calculation of automobile miles per gallon of gasoline in low-track classes or the application of deductive logic learned in geometry to the proof of theorems and corollaries in calculus.

It is clear that both the knowledge presented and the intellectual processes cultivated in English classes and the access to mathematical content in math classes were quite different at different track levels. Moreover, these differences seem to be more than simply a result of accommodating individual needs—a major reason given for such curricular variation. The types of differences found indicate that, whatever the motives for them, social and educational consequences for students are likely to flow from them. The knowledge to which different groups of students had access differed strikingly in both educationally and socially important ways.

Much of the curricular content of low-track classes was such that it would be likely to lock students into that track level—not so much as a result of the topics that were included for instruction but because of the topics that were omitted. Many of the topics taught almost exclusively to students in low-track classes may be desirable learnings for all students—consumer math skills, for example. But these topics were taught to the exclusion of others—introduction to algebraic equations, for example—that constitute prerequisite knowledge and skills for access to classes in different, and higher, track levels. So, by the omission of certain content from low-track classes, students in effect were denied the opportunity to learn material essential for mobility among track levels. This content differentiation was found as early as grade six. Contrary to the suspicions of many, this line of thinking, however, does not imply that all students need the same things in school. Moreover, it is not in conflict with the view that schools should accommodate differences among individuals in learning speed and style, nor does it deny that some stu-

dents need remediation in fundamental prerequisite skills. But it suggests, given the importance of some curricular topics for students' future educational opportunities, that individualization and remediation should take place within the context of a core of educationally and socially important learnings—thus at least providing equal access to these topics.

We make certain assumptions about what different kinds of students are able to learn. This extends not only to what we believe they are able to master in the way of skills but also to what ideas and concepts we believe are appropriate for them to be exposed to. As a result, a student with poor reading skills may never be exposed to a Shakespearean play—literature highly valued, if not widely read, in our culture. A teacher of students with low-level reading skills may be justified in not expecting these students to read with understanding the text of a Shakespeare play. On the other hand, exposure to these works seems entirely appropriate. Franco Zeffirelli's film version of *Romeo and Juliet*, for example, is not only suitable for but fascinating to teenage students of every ability and achievement level. The cultural context of the work, the themes represented in the play, and examples of the often rather base comedy used as relief from the intensity of the play's tragic events are valuable and interesting topics for all English students—even for those perceived to be at the lowest ability levels, even for those not expected to go on to colleges and universities. That Shakespeare, originally written to appeal to the Elizabethan masses—far less literate than even our least skilled American teenagers—is now reserved only for the elite of our schools, those identified as having the most intellectual promise, would be silly, if it were not for the serious consequences of differentiated content for students.

Learning Beyond Content

QUESTION: What are the five most critical things you want the students in your class to learn this year? By learn, we mean everything that the student should have upon leaving the class that (s)he did not have upon entering.

We asked this question of each of the teachers during our interviews with them. Among their responses, teachers of high-track classes typically included the following kinds of answers.

RESPONSES: Interpreting and identifying.
Evaluation, investigating power
High-track Science—junior high

Deal with thinking activities—Think for basic answers—es-
say-type questions.
High-track English—junior high

Ability to reason logically in all subject areas.
High-track math—senior high

The art of research.
High-track English—senior high

Learn how to test and prove ideas. Use and work with sci-
entific equipment. Learning basic scientific facts and prin-
ciples.
High-track Science—junior high

Scientific reasoning and logic.
High-track Science—senior high

Investigating technology, investigating values.
High-track Social Science—junior high

Self-reliance, taking on responsibilities themselves.
High-track Science—junior high

To learn values and morals—to make own personal deci-
sions.
High-track English—junior high

To think critically—to analyze, *ask* questions.
High-track Social Science—junior high

Individual interpretation of materials covered.
High-track English—senior high

Logical thought processes. Analysis of given information.
Ability to understand exactly what is asked in a question.
Ability to perceive the relationship between information that
is given in a problem in a statement and what is asked.
High-track Science—senior high

Love and respect for math—want them to stay curious, ex-
cited and to keep believing they can do it.
High-track Math—junior high

To realize that all people are entitled to certain inalienable
rights.
High-track Social Science—junior high

To think critically (analyzing).
High-track English—senior high

How to think critically—analyze data, convert word problems into numerical order.
High-track Math—senior high

To be creative—able to express oneself.
High-track English—senior high

The most important thing—think more logically when they leave.
High-track Math—senior high

Ability to think and use information.
Concept development.
High-track Science—senior high

Ability to think for themselves.
High-track Science—senior high

How to evaluate—think objectively. To think logically and with clarity and to put it on paper. To be able to appreciate a variety of authors' works and opinions without judging them by their own personal standards.
High-track English—senior high

Confidence in their own thoughts.
High-track English—senior high

Able to collect and organize information. Able to think critically.
High-track Social Science—junior high

Determine best approach to problem solving. Recognize different approaches.
High-track Math—senior high

That their own talents and thoughts are important. Development of imagination. Critical thinking.
High-track English—senior high

Problem-solving situations—made to think for themselves. Realizing importance of their education and use of time. Easy way is not always the best way.
High-track Science—senior high

Better feeling for their own abilities and sense of what it's like in a college course.
High-track Math—senior high

To gain some interpretive skills.
High-track English—senior high

Teachers of low-track classes said the following kinds of learnings were essential for their students:

Develop more self-discipline—better use of time.
Low-track English—junior high

Respect for each other
Low-track Math—junior high

I want them to respect my position—if they'll get this, I'll be happy.
Low-track Math—junior high

That they know that their paychecks will be correct when they receive them. Punctuality, self-discipline and honesty will make them successful in their job. They must begin and end each day with a smile. To be able to figure their own income tax (at the) end of the year. Properly planning to insure favorable performances.
Low-track Math—senior high

Self-discipline, cooperativeness, and responsibility.
Low-track Science—junior high

I teach personal hygiene—to try to get the students to at least be aware of how to keep themselves clean.
Low-track Vocational Education—junior high

Independence—start and complete a task on their own.
Low-track English—senior high

Responsibility of working with people without standing over them.
Low-track Science—senior high

Ability to use reading as a tool—e.g., how to fill out forms, write a check, get a job.
Low-track English—junior high

How to fill out insurance forms. Income tax returns.
Low-track Math—senior high

Understanding the basic words to survive in a job. Being able to take care of their own finances—e.g., banking, income tax, etc. Being able to prepare for, seek and maintain a job. To associate words with a particular job.
Low-track English—senior high

To be able to work with other students. To be able to work alone. To be able to follow directions.
Low-track English—junior high

Socialization—retarded in social skills.
Low-track English—junior high

How to cope with frustration.
Low-track English—junior high

Business-oriented skills—how to fill out a job application.
Low-track English—junior high

More mature behavior (less outspoken).
Low-track Science—junior high

Respect for their fellow man (students and teachers).
Low-track Science—junior high

Learn to work independently—use a sense of responsibility.
Low-track Science—senior high

Content—minimal. Realistic about goals. Develop ones they can achieve.
Low-track Science—senior high

Practical math skills for everyday living. A sense of responsibility.
Low-track Math—senior high

To learn how to follow one set of directions at a time, take a directive order and act upon it.
Low-track Social Science—junior high

Life skills. Work with checking account.
Low-track Math—junior high

Good work habits.
Low-track Math—junior high

Respect. Growth in maturity.
Low-track Math—junior high

It is obvious from these answers that among those learning outcomes teachers desire for their students are attitudes and behaviors not specifically related to the content areas. This is not surprising since, historically, our society has asked schools to do far more than provide academic learnings for students. Schools have been expected to transform the young into contributing members of society, imbue them with a strong commitment to our cultural ideals, increase their skills in interpersonal relationships, and contribute to their physical and mental well-being, to name only a few of their responsibilities.⁵

Knowing this, when we asked teachers during interviews to name the five most critical things they wanted their students to learn during their year with them, we expected that they would list nonacademic as

well as academic learning goals. As we have just seen, many of them did. Among the senior high English classes we studied, thirty-seven teachers (45 percent) mentioned these non-subject-related behaviors as instructional goals or content and twenty-six teachers (35 percent) of junior high-middle school English classes included these types of learnings. Similar percentages of the math teachers included non-subject-related behaviors as topics of instruction or as desired learnings. At the senior high level, thirty-five teachers (49 percent) and at the junior high level, twenty-eight teachers (41 percent) included these behaviors as instructional content. Throughout the discussion of these responses it should be borne in mind that they represent only about half of the high school classes and about a third of the junior high-middle school classes we studied.

Two types of nonacademic learnings were mentioned most frequently by these teachers—one concerning desired student behaviors in the area of personal deportment, the other regarding desired behaviors considered part of the learning process or necessary to carry out class procedures. We analyzed these statements first by classifying them into one of the following categories: (1) statements indicating that the teacher was seeking student autonomy and independence (e.g., “confidence in own thoughts”); (2) statements indicating that the teacher encouraged student conformity to teacher authority and established classroom routines (e.g., “learn to follow directions accurately and promptly”); and (3) statements indicating that both types of behaviors were encouraged or statements difficult to interpret as distinctly belonging to either of the previous two categories (e.g., “self-discipline” was a response that could mean either independent or conforming behavior).

The two lists below give some examples of the kinds of learning teachers hoped for that were classified into the first two categories:

1. *independence*
 - critical thinking
 - work on individual projects or assignments
 - self-direction
 - creativity
2. *conformity*
 - getting along with others
 - working quietly
 - improving study habits
 - punctuality—both in attendance and handing in assignments

cooperation
conforming to rules and expectations

We looked then at the ways in which these non-subject-specific learning goals were distributed among track levels. We found in most of the classes we studied that students in different track levels were expected to learn different kinds of behaviors that were not actually related to the subjects they were studying.

Among the English classes, both the junior and senior high school teachers who emphasized non-subject-related learnings expected different kinds of behaviors from their high- and low-track students. We found that teachers of high-track classes were more likely to emphasize such behaviors as critical thinking, independent work, active participation, self-direction, and creativity than were other teachers. At the same time, teachers of low-track classes were more likely than others to emphasize student conformity: students getting along with one another, working quietly, improving study habits, being punctual, and conforming to classroom rules and expectations. This pattern was also found among senior high math teachers. At the junior high level, math teachers did not systematically differ in their behavioral expectations of students. From the responses of these teachers to our question, it seems that most of them wanted both independence and compliance from their students rather than strongly emphasizing one over the other. Average classes were closer to the high than to the low tracks in this respect as well as in content. Teachers of average groups (except in junior high math) tended to have “independent” student behaviors as their goals far more than did teachers of low groups. The low groups stood apart from the other kinds of classes in their teachers’ emphasis on conformity and passive compliance.

Apparently many teachers were able to effect this learning in their students. At least we know that many students reported non-subject-related behaviors as the most important things they learned or did in class. And, as is clearly seen in the following responses, the pattern of differences found in teachers’ learning goals for students at different track levels are also reflected in what students say they learned.

QUESTION: What is the most important thing you have learned or done so far in this class?

High-track students named the following kinds of things:

RESPONSES: I have learned to form my own opinions on situations. I have also learned to not be swayed so much by another person's opinion but to look at both opinions with an open mind. I know now that to have a good solid opinion on a subject I must have facts to support my opinion. Decisions in later life will probably be made easier because of this.

High-track English—senior high

I've learned to study completely, and to know everything there is to know.

High-track English—senior high

I have learned to speak in front of a group of people, and not be scared to death of everyone.

High-track English—senior high

To know how to communicate with my teachers like friends and as teachers at the same time. To have confidence in myself other than my skills and class work.

High-track English—junior high

I have learned to be creative and free in doing things.

High-track English—senior high

I have learned how to make hard problems easier to solve.

High-track Math—senior high

The most important thing I have learned in this class this quarter is how to express my feelings.

High-track English—senior high

How to organize myself and present an argument.

High-track English—senior high

I'm learning how to communicate with large groups of people.

High-track English—senior high

The most important thing I have learned or done in this class is I now have the ability to be able to speak in front of a crowd without being petrified, as I was before taking this class.

High-track English—senior high

I want to be a lawyer and debate has taught me to dig for answers and get involved. I can express myself.

High-track English—senior high

The most important thing I have learned is how to speak in front of a group of people with confidence.

High-track English—senior high

I have learned how to argue in a calm and collected way.

High-track English—senior high

How to express myself through writing and being able to compose the different thoughts in a logical manner; this is also a class where I may express my creativity.

High-track English—senior high

Learned to think things out. Like in a book I learned to try and understand what the author is really saying to find the author's true thoughts.

High-track English—senior high

I've learned to look into depth of certain things and express my thoughts on paper.

High-track English—senior high

The most important thing I have learned in this class is to loosen up my mind when it comes to writing. I have learned to be more imaginative.

High-track English—senior high

How to present myself orally and how to listen and to think quick.

High-track English—senior high

To understand concepts and ideas and experiment with them. Also, to work independently.

High-track Science—senior high

My instructor has opened a whole new world for me in this class, I truly enjoy this class. He has given me the drive to search and find out answers to questions. If there is one thing that I have learned from this class it would have to be the want for learning.

High-track Social Studies—senior high

It taught us how to think in a logical way to work things out with a process of elimination.

High-track Math—senior high

I have learned that in high school the English classes treat you more like an adult.

High-track English—senior high

I have proved to myself that I have the discipline to take a difficult class just for the knowledge, even though it has nothing to do with my career plans.

High-track Math—senior high

I have learned that I have a wider span of imagination than

I thought. I have also learned to put how I feel and what I feel into words and explain them better.

High-track English—senior high

Many times in this class I get wrong answers, but in this class you learn to learn from your mistakes, also that even if you do have a wrong answer you should keep trying and striving for that correct answer. This, along with the subject I have learned from this class, and I think it's very important.

High-track Math—senior high

How to think and reason logically and scientifically.

High-track Math—senior high

I think the most important thing I've done in this class is exercise my brain. To work out problems logically so I can learn to work out problems later in life logically.

High-track Math—senior high

Brains work faster and faster.

High-track Math—senior high

Learning about how others respond and act—what makes them do the things they do—talking about relations, and how happenings in earlier life can affect children when they are growing up. The class discussions make it really interesting and the teacher startles us sometimes because he can really understand things from our point of view.

High-track Social Science—senior high

The most important thing that I have learned in this class is the benefit of logical and organized thinking, learning is made much easier when the simple processes of organizing thoughts have been grasped.

High-track Math—senior high

Low-track students were more likely to give answers like these:

Behave in class.

Low-track English—junior high

I have learned that I should do my questions for the book when he asks me to.

Low-track Science—senior high

Self-control.

Low-track Social Studies—junior high

Manners.

Low-track English—junior high

How to shut up.

Low-track Vocational Education—junior high

The most important thing I have learned in this class is to always have your homework in and have materials ready whenever she is ready.

Low-track Vocational Education—senior high

Write and getting my homework done.

Low-track English—junior high

Working on my Ps and Qs.

Low-track English—junior high

I think the most important is coming into class and getting our folders and going to work.

Low-track Math—junior high

I have learned about many things like having good manners, respecting other people, not talking when the teacher is talking.

Low-track English—junior high

Learned to work myself.

Low-track Math—junior high

I learned about being quiet when the teacher is talking.

Low-track Social Studies—junior high

To learn how to listen and follow the directions of the teacher.

Low-track Math—senior high

I learn to respect the teacher.

Low-track Vocational Education—junior high

Learn to get along with the students and the teacher.

Low-track English—junior high

How to go through a cart and find a folder by myself.

Low-track Math—junior high

To be a better listener in class.

Low-track English—senior high

In this class, I have learned manners.

Low-track English—junior high

What are the implications of these findings? Unfortunately, the data themselves don't tell us. But it seems important to consider the possible meanings of these trends. Why would teachers of adolescents considered to be of low ability or achievement consider learning to be "less outspoken," or to "cope with frustration," or to "keep themselves clean,"

or to "begin and end each day with a smile," or to "take a directive order and act upon it" as among the five most *critical* things students should learn during the year? Of course, not every low-track teacher said these kinds of things, but many did. And further, teachers of high-track classes tended *not* to have these kinds of goals for their students.

Are these behaviors that we know are necessary for remediating deficiencies in student achievement? That is, by acquiring these behaviors, are "slow" students more likely to catch up with their average or most able peers? I know of no evidence that suggests that these behaviors enhance students' dispositions to learn or their ability to do so.

Why, then, are these behaviors desired for slow students? And, also important, why are they *not* among the learning goals for bright ones? Do smart students already possess compliant and passive behavior patterns? If so, are these attributes a part of what makes them appear smart to school people? This also seems unlikely; we do not generally judge passive, compliant, "not outspoken" people to be bright *because* of these characteristics. And even if it were the case, why would teachers of average and especially high-achieving groups of students want what seems to be almost the opposite type of behaviors to be learned by their classes? Why instead of passivity do such teachers seek independence and autonomy in their students? Why do they hope for creativity and critical thinking?

Some possibilities come to mind. First, we might reasonably speculate that low-track students are perceived as unmanageable by their teachers. The thought of thirty or more potentially unmanageable students together in a classroom can certainly lead teachers to value control over anything else, even academic learning. And it seems clear that the behavioral goals for low classes are more closely linked to control than they are to learning. Of course, learning course content is certainly more likely in a situation that is not characterized by chaos. But are slow students in fact characteristically unmanageable? Does classroom misbehavior constitute a part of low ability or achievement? Or, and this seems more likely, are unmanageable students often labeled slow and placed in low-track classes regardless of their academic aptitude? Schools make very few efforts to distinguish between students who are poor achievers because they lack skills or even aptitude and those who are low achievers because of disruptive behavior or poor attendance. What happens, it seems, is that we may saddle those who find it hard to learn with those who find it hard to behave. What does this possibility say about the chances

for slow students to learn in class? And, further, doesn't it follow that the reason high-track classes almost never have to concern themselves with issues of control may be that disruptive students are almost always selected out and sent to low classes, regardless of their "ability"?

This prospect leads us to another frightening possibility. If teachers and schools tend to associate misbehavior and nonconformity with "slowness" and place students accordingly, are these perceptions more often made about poor and minority students? We know that these students are more often found in low tracks. Could this be partly because they are perceived as less easily controlled, less conforming, and more in need of learning passivity and compliance? Could some of teachers' fears, if indeed they are at work here, be based on racial, cultural, or social-class differences? These are questions that need to be considered seriously.

A third possibility has to do with the schools' perceived mission to prepare the young for roles in adult society. Part of what students are taught in school is a set of behaviors that will make them suitable for the places they will hold in society as adults. Is it possible that the non-subject-related behaviors teachers have as learning goals for their students are the kinds of behaviors they think the students will need as adults? If this is the case, even if only partly so, the implications are serious for all students. And when we consider how students end up sorted by race and class in schools, the consequences for the futures of poor and minority students may be especially devastating.

Could it be that we are teaching kids at the bottom of the educational hierarchy—who are more likely to be from poor and minority groups—behaviors that will prepare them to fit in at the lowest levels of the social and economic hierarchy? And, at the other extreme, are we teaching kids at the top of the schooling stratification system behaviors that are most appropriate for professional and leadership roles? In essence, are we teaching kids at the bottom how to stay there and kids at the top how to get ahead?

When we look at the differences in these non-subject-related learning goals next to the differences in curricular content, we see similar patterns. Students are being exposed to knowledge and taught behaviors that differ not only educationally but in socially important ways. Students at the top are exposed to the knowledge that is highly valued in our culture, knowledge that identifies its possessors as "educated." This knowledge, too, is that which permits access to certain educational fu-

tures—specifically, college or university attendance. And, of course, the credentials resulting from higher education are required for access to upper-level social and economic positions in the adult world. Similarly, the non-subject-related behaviors that teachers have as learning goals for high-track students are those that are highly valued in our society. Behavior patterns such as independent and critical thinking, creativity, and questioning have a certain status in our culture. We see these as attributes of leaders, of decision makers; in short, we see them as qualities we admire in smart people at the top.

The knowledge and behaviors taught to low-track students are, for the most part, of another sort altogether. It's not that they are not *valued* at all but rather that they have little exchange value in a social or economic sense; they have no prestige, nor do they permit special access. Everyone, for example, needs to acquire functional literacy and computation skills, not so much for the benefit of the individual as for the functioning of society as a whole. Students who have access only to this low-prestige knowledge are being denied, by omission, access to both educationally and socially important learnings. Further, the non-subject-related behaviors that teachers expect low-track students to learn are of a similar type. While necessary to keep the social system functioning smoothly, behavior patterns such as following directions, good work habits, punctuality, being realistic about goals, and so forth are not those attributes we most admire in our leaders. They are, however, those we demand from people in lower-level occupations. Teachers say they believe tracking helps them better meet the individual needs of students. Are these the kinds of things they see as differing needs? I expect not. It is likely that teachers *intend* to adapt instruction to differences in students' learning styles and learning speeds. What they *effect* with differentiated curricular content, however, appears to be of another sort altogether.

This, of course, is somewhat speculative. But, at the same time, we would be foolish to ignore these possibilities. Whenever we sort students in schools and treat them differently, we need to examine all the possible effects of these practices. We cannot merely assume that strictly educative mechanisms and purposes are being exercised. The possibility that what goes on in classes at different track levels may have different social and economic effects is a theme we will return to repeatedly as we continue to explore these 299 classes.

5

Opportunities to Learn

Even if teenagers in classes at different track levels are exposed to identical curricular contents, differences in classroom conditions can create important differences in what they learn. Most people would readily agree that what knowledge students are exposed to is important in determining the quality of education they obtain. But probably as critical in the access students have to learning in schools are the conditions they encounter there. School and classroom situations can either enhance or restrict the chances that students will learn whatever content is presented. I am convinced, in fact, that certain critical features of the schooling context may actually constitute students' opportunities to learn.¹

In this chapter we will look carefully at two of those classroom conditions—time to learn and quality of instruction—that appear to influence strongly whether student learning is likely to occur. We will also examine how these conditions differed in our sample of tracked math and English classes. The differences we found were marked—enough so that we believe they provided students in various track levels with greater and lesser learning opportunities.² And given the vast content differences we found between high- and low-track classes, these classroom differences served to further widen the gap between the prospects of various groups of students—their chances to get the best education our schools had to offer.

Before looking at these school-based differences, however, we must clarify two important premises underlying our interpretation of what we

quoted in D. A. Cohen and M. K. Lazerson, "Education and the Corporate Order," *Socialist Revolution*, 1972, 2, p. 53.

38. Boston, *School Documents*, 1908, no. 7; p. 53, cited in Lazerson, *Origins of the Urban School*, p. 189.

39. *Report to the Commission on National Aid to Vocational Education together with the Hearings Held on the Subject*, U.S. House of Representatives, 63rd Cong., 2d Sess., Doc. no. 0004 (Washington, D.C.: U.S. Government Printing Office, 1914), 1:12, as quoted in Becker, "Education and Work," p. 3.

40. Cited in Spring, *Education and the Rise of the Corporate State*, p. 111.

41. W. B. Pillsbury, "Selection—An Unnoticed Function of Education," *Scientific Monthly*, January 1921, 12, p. 71, as quoted in Cohen and Lazerson, "Education and the Corporate Order."

42. Cited in S. Bowles and H. Gintis, *Schooling in Capitalist America* (New York: Basic Books, 1976), p. 192.

43. L. Terman, *Intelligence Tests and School Reorganization* (New York: World Book Co., 1923), pp. 27-28.

44. *Ibid.*, as cited in Bowles and Gintis, *Schooling in Capitalist America*, p. 123.

45. C. J. Karier, "Elite Views on American Education," in *Foundations of Education: Dissenting Views*, ed. J. J. Shields and C. Geer (New York: Wiley, 1974), p. 47.

46. Cohen and Lazerson, *Education and the Corporate Order*, p. 53.

47. An excellent discussion of the conceptualization of the idea of measurable intelligence and the social and political uses of this idea can be found in S. J. Gould, *The Mismeasure of Man* (New York: W. W. Norton, 1981).

48. Charles A. Prosser, as cited in Kliebard, "The Drive for Curriculum Change," p. 279.

49. J. Conant, *The American High School* (New York: McGraw-Hill, 1959).

50. H. M. Kliebard, "The Drive for Curriculum Change," makes this point citing J. D. Koerner in his article "The Tragedy of the Conant Report," *Phi Delta Kappa*, 1960, 42, 121 as a "lone voice" of criticism.

Chapter 3

1. For more detailed information about A Study of Schooling generally and more specifics about the methodology used, see the final report of the study, John I. Goodlad, *A Place Called School* (New York: McGraw-Hill, 1981); the following technical reports of the study will be helpful: J. I. Goodlad et al., *A Study of Schooling: Series of Introductory Descriptions* (1979-1980), Technical Report no. 1; and B. C. Overman, *A Study of Schooling: Method-*

ology, Technical Report no. 2, both available from the ERIC clearinghouse on teacher education.

2. *Ibid.*

3. W. Findley and M. Bryan, M. *The Pros and Cons of Ability Grouping* (Washington, D.C.: National Education Association, 1975).

Chapter 4

1. Administrators at the Newport schools prohibited the collection of race or ethnicity data about individual students. They did, however, provide us with the percentages of different groups that constituted their schools' populations.

2. The analysis of the distribution of white and nonwhite students into high- and low-track classes in six multiracial schools yielded a chi-square significant at the .001 level with 1 df. See J. Oakes, *A Question of Access: Tracking and Curriculum Differentiation in a National Sample of English and Mathematics Classes*, A Study of Schooling Technical Report no. 24 (Los Angeles: University of California, 1981), available from the ERIC clearinghouse on teacher education, for a complete presentation of this analysis.

3. The findings presented here are the results of discriminant analyses conducted separately for each construct in each subject area at each of the two levels of schooling. For a detailed presentation of these analyses and precise definitions of the variables and summary statistics, see the report cited above.

4. J. H. Spring, *The Sorting Machine* (New York: David McKay, 1976).

5. J. I. Goodlad, *What Schools Are For* (Bloomington, Ind.: Phi Delta Kappa, 1979).

Chapter 5

1. The term *opportunity to learn* has come to mean different things to various educational researchers. Carroll, for example, in his model for school learning, used the term to mean the specific content covered during instruction (J. B. Carroll, "A Model for School Learning," *Teachers College Record*, 1963, 64, 723-33).

2. See R. Barr and R. Dreeken, *How Schools Work* (Chicago: University of Chicago Press, 1983), for an excellent discussion of the interrelatedness of group composition, variation in instructional conditions, and student learning outcomes at the elementary level.

3. For example, see J. Stallings, M. Needels, and N. Stayrook, *The Teaching of Basic Reading Skills in Secondary Schools, Phase II and Phase III* (Menlo Park, Calif.: SRI International, 1979).

4. D. Wiley and A. Harnischfeger, "Explosion of a Myth: Quantity of